EXCELLENCE

CLASS 11 - 2026-27

Physics | Motion 03

Motion in a Straight Line

New problems · Cube & paths · Speed and average velocity

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What is new today

01 Cube: fly through air vs crawl on faces

02 Carrom path by the mirror method

03 Drunkard steps and a rolling wheel

04 x(t): catch the turning point inside the time

05 Speed, average speed, average velocity

06 Two average-speed recipes and their uses

What we will not redo

Motion 01 and 02 already fixed frames, distance versus displacement, polygons, circular arcs, and how to read an x–t graph. Those pages stay closed today.

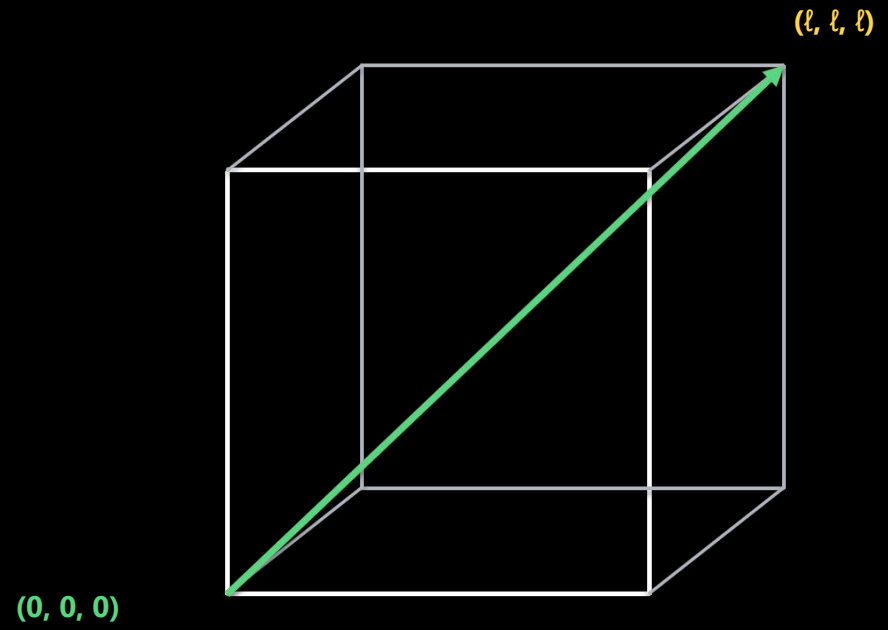
Today starts where those lectures stop: shortest path on a surface, a reflected carrom shot, a point on a rolling wheel, motion given by x(t), and the first speed–velocity formulas.

Butterfly through a cube

A butterfly can fly. It starts at one corner of a cubical room of side ℓ and reaches the opposite corner. The flight path is unknown, so distance cannot be found. Displacement is the body diagonal.

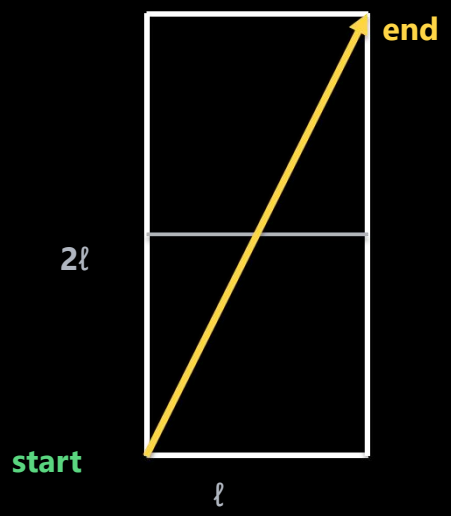
$$|\Delta \mathbf{r}| = \sqrt{(\ell^2 + \ell^2 + \ell^2)} = \ell\sqrt{3}$$

Displacement = $\ell\sqrt{3}$ · distance unknown



Ant on the faces — shortest crawl

An ant cannot fly. It must stay on the walls. Displacement is still the body diagonal $\ell\sqrt{3}$. The shortest distance is found by opening two faces into one flat rectangle and drawing a straight line.



min distance = $\sqrt{((2\ell)^2 + \ell^2)} = \ell\sqrt{5}$

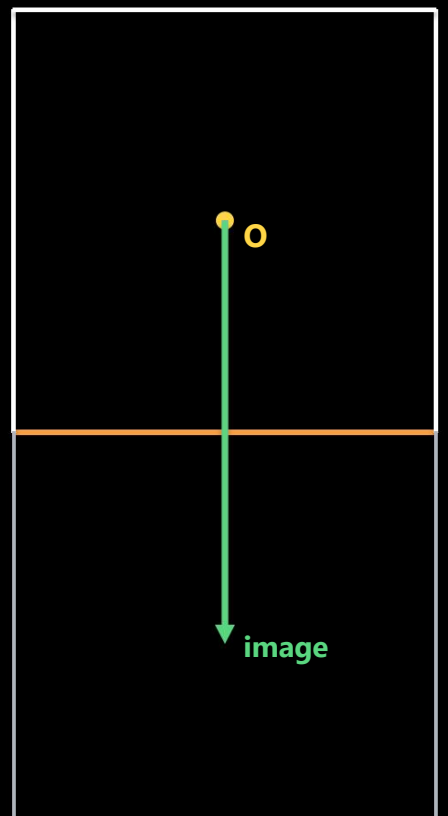
displacement stays $\ell\sqrt{3}$

3ℓ along three edges is a possible crawl, not the shortest.

Rough work / extra work

Carrom — mirror the board

A 4 ft × 4 ft board. The queen starts at the centre, hits the front edge, and falls into the hole behind the striking line. Reflect the board across that edge. The straight line on the double board is the real path length.



path = $\sqrt{(6^2 + 2^2)} = \sqrt{40} = 2\sqrt{10}$ ft

displacement = centre to corner = $2\sqrt{2}$ ft

distance $2\sqrt{10}$ ft · |displacement| $2\sqrt{2}$ ft

Drunkard and the pit

He takes 5 steps forward and 3 steps back, again and again. Each step is 1 m and takes 1 s. A pit is 11 m ahead.

One cycle: +5 then −3 → net +2 m in 8 s

Three cycles: 6 m in 24 s. He is still short of the pit.

Next 5 forward steps: 6 + 5 = 11 m. He falls in. No return.

Time = 24 + 5 = 29 s

Do not add a backward 3 after he has already reached 11 m.

MCQ — time to the pit

The drunkard falls into the pit after

1. 21 s
2. 29 s
3. 31 s
4. 37 s

Solution: MCQ — time to the pit

Step 1: $8 + 8 + 8 + 5 = 29$ s

Answer: 29 s

Point on a pure-rolling wheel

A wheel of radius R rolls without slipping. Point P starts at the bottom, touching the road. We want the displacement of P, not the path length of P.



Half turn

$|\Delta r| = \sqrt{((\pi R)^2 + (2R)^2)}$

$= R\sqrt{(\pi^2 + 4)}$

Full turn brings P back to the road, but the wheel has moved forward.

Rough work / extra work

Position given by x(t)

When the question gives x as a formula in t, do not assume the object keeps one direction. Find the instant when velocity is zero. If that instant lies inside the asked time, the object turned there.

1. **v = dx/dt. Set v = 0 and solve for t.**
2. **Keep a root only if it lies between the start and end times.**
3. **Displacement = x(end) − x(start).**
4. **Distance = sum of |Δx| on each piece split by the turn.**

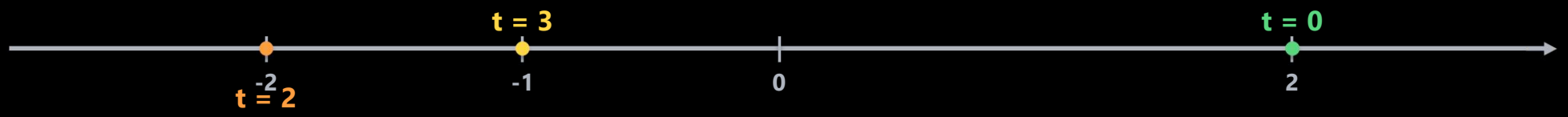
$x = t^2 - 4t + 2$, from 0 to 3 s

$v = 2t - 4 = 0 \rightarrow t = 2\ s$ (inside 0 to 3)

$x(0) = 2$, $x(2) = -2$, $x(3) = -1$

displacement $= -1 - 2 = -3$ m

distance $= |-2 - 2| + |-1 - (-2)| = 4 + 1 = 5$ m



$x = -t^2 + 6t + 4$, from 0 to 4 s

$v = -2t + 6 = 0 \rightarrow t = 3$ s (inside 0 to 4)

$x(0) = 4$, $x(3) = 13$, $x(4) = 12$

displacement $= 12 - 4 = 8$ m

distance $= |13 - 4| + |12 - 13| = 9 + 1 = 10$ m

Turn inside the interval → distance is larger than |displacement|

Arc plus two closing sides

Radius R = 21 cm. The object travels a 240° arc, then two straight pieces of length R each, and the loop closes.

arc = (240/360) × 2πR = (2/3) × 2π × 21

Use π = 22/7: arc = 88 cm

two straight pieces = 21 + 21 = 42 cm

distance = 88 + 42 = 130 cm

Same ends, two different routes

Both travellers start and finish at opposite ends of a diameter. A walks the semicircle. B walks three sides of the inscribed regular hexagon. Side of the hexagon equals R.

|displacement| is the same for both: the diameter 2R. Ratio = 1.

distance of A = πR , distance of B = 3R

distance ratio A : B = πR : 3R = π : 3

Rough work / extra work

Speed and velocity — first definitions

Speed uses distance. Velocity uses displacement. That is the only split you need before the formulas.

average speed = total distance / total time (scalar)

average velocity = total displacement / total time (vector)

instantaneous speed = |dx/dt| (size of the slope of x–t)

Two ways to mix speeds

If each piece is given by a speed and a time, weight by time. If each piece is given by a speed and a distance, weight by distance. Do not average the two speeds by adding and dividing by 2 unless the × are equal.

speeds for times: $(s_1\,t_1 + s_2\,t_2) / (t_1 + t_2)$

speeds for distances: $(x_1 + x_2) / (x_1/s_1 + x_2/s_2)$

Equal × → ordinary mean. Equal distances → the distance formula, not $(s_1+s_2)/2$.

Four straight trips — one table

Trip	avg speed	|avg velocity|
0 → 20 m in 4 s	5 m/s	5 m/s
0 → 20 → 5 m in 10 s	3.5 m/s	0.5 m/s
0 → −20 → 10 m in 5 s	10 m/s	2 m/s
out and back, 40 m path, 10 s	4 m/s	0

Path length goes in the speed. Only start-to-end goes in the velocity.

Semicircle in 10 s

Radius 4 m. The object runs the semicircle from one end of the diameter to the other in 10 s.

distance = πR = 4π m , |displacement| = 8 m

average speed = 4π / 10 = 0.4 π m/s

|average velocity| = 8 / 10 = 0.8 m/s

Constant speed v on a circle

One lap takes time $2\pi R/v$. Read each fraction of the lap from that.

Part	distance	|disp|	$\|v_avg\|$
1/4 lap	$\pi R/2$	$R\sqrt{2}$	$(R\sqrt{2}) / t$
1/2 lap	πR	$2R$	$2v/\pi$
full lap	$2\pi R$	0	0

Average speed for a full lap is still v. Average velocity is zero.

Rough work / extra work

Half the distance at v, half at 2v

time = (d/2)/v + (d/2)/(2v) = 3d / (4v)

average speed = d / (3d/(4v)) = 4v/3

Same direction the whole way, so |average velocity| equals this average speed.

Answer: 4v/3

Out at 30 km/h, back at 60 km/h

Same road both ways. Distances are equal, so use the distance formula. Let the one-way distance be d.

average speed = 2d / (d/30 + d/60) = 2 / (1/30 + 1/60)

= 2 / (3/60) = 40 km/h

Round trip of two speeds: $2\,s_1\,s_2 / (s_1 + s_2)$

Displacement for the round trip is zero, so average velocity is zero.

MCQ — three equal parts

First third at 10 m/s, second at 20 m/s, last at 30 m/s, all same direction. Average velocity is

1. 17.8 m/s
2. 16.4 m/s
3. 18.3 m/s
4. 20.2 m/s

Solution: MCQ — three equal parts

Step 1: $t = (x/10) + (x/20) + (x/30) = 11x/60$

Step 2: total distance $= 3x$, average $= 3x / (11x/60) = 180/11$

$180/11 \approx 16.4$ m/s

Unequal times, then a mixed half

v_1 for time t/3 and v_2 for time 2t/3

average = $(v_1 + 2\,v_2) / 3$

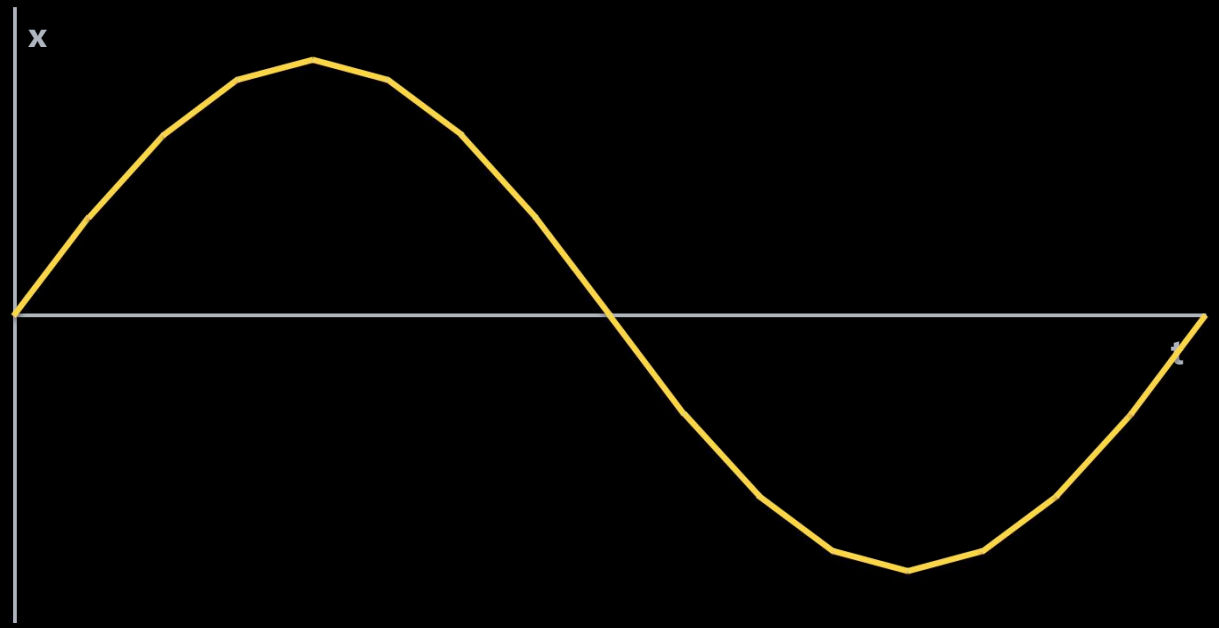
A different mix: first half of the distance at 5 m/s, second half at 15 m/s.

average = 2×5×15 / (5+15) = 7.5 m/s

Rough work / extra work

One full wave on x–t

Position swings between +10 m and −10 m and returns to the start after 8 s. That is one full oscillation.



distance ≈ 4 × 10 = 40 m in 8 s

average speed = 40/8 = 5 m/s

displacement = 0 → average velocity = 0

Only today’s lines

- Fly the cube: |displacement| = $\ell\sqrt{3}$. Crawl it: shortest distance = $\ell\sqrt{5}$.
- A bounce becomes a straight line after you reflect the board.
- Drunkard: count full cycles, then the last forward run into the pit.
- Rolling point, half turn: $R\sqrt{(\pi^2 + 4)}$.
- In x(t), split the path only at a turning time inside the interval.
- Equal times: ordinary mean. Equal distances: the distance formula.

Next lecture continues velocity and acceleration. Today’s formulas stay.

Thank you

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